

Proof ledger

Repository source: proof/PROOF_LEDGER.md

Status vocabulary

- **ESTABLISHED IN MANUSCRIPT:** a complete derivation is present in the paper, subject to ordinary mathematical review.
- **EXACT-CERTIFIED:** the finite claim is checked with integer and rational arithmetic by the standalone verifier.
- **NUMERICAL AUDIT:** a separate floating-point implementation agrees.
- **RECONSTRUCTION-CHECKED:** an expected derived artifact is rebuilt byte for byte.
- **MUTATION-TESTED:** a targeted corrupted object is rejected.
- **SOURCE-CHECKED:** bibliographic or historical wording has been compared with an identified source.
- **RELEASE-PENDING:** technically prepared but awaiting metadata, hosted workflow, tag, or explicit approval.
- **EXTERNAL REVIEW NEEDED:** no independent subject-matter review has been recorded.

Mathematical claims

ID	Claim	Evidence	Status
M1	Constant-width support identity and diameter-one normalization	Paper Section 2; classical references	ESTABLISHED IN MANUSCRIPT; SOURCE-CHECKED
M2	Blaschke volume-surface identity	Paper Section 2; classical references	ESTABLISHED IN MANUSCRIPT; SOURCE-CHECKED
M3	Minimum circumsphere contact balance and Jung range	Paper Section 2	ESTABLISHED IN MANUSCRIPT
M4	Opposite-point identity and radial support inequality	Paper Lemma 2.1	ESTABLISHED IN MANUSCRIPT
L1	Transport inequality and tangent-gap master inequality	Paper Section 3	ESTABLISHED IN MANUSCRIPT
L2	Rational propagation implications	Paper Section 3; exact lower-cell checks	ESTABLISHED IN MANUSCRIPT; EXACT-CERTIFIED
L3	Four refined cells replace 03A1 with positive margins	Refined NPZ cells and JSON	EXACT-CERTIFIED; NUMERICAL AUDIT
L4	Complete lower partition is gap free and has unique bottleneck 03B02	Final JSON and exact verifier	EXACT-CERTIFIED; RECONSTRUCTION-CHECKED
N1	Near-Jung four-contact deficiency budget	Paper Section 4	ESTABLISHED IN MANUSCRIPT
N2	Affine contact coordinates and Gram formulas	Paper Section 4	ESTABLISHED IN MANUSCRIPT

ID	Claim	Evidence	Status
N3	Determinant transfer $\det H \geq 1 - S/2 - S^2/2$	Paper Proposition 4.1	ESTABLISHED IN MANUSCRIPT; exact endpoint evaluation
N4	Dominant-edge coordinate-width diagonal bounds	Paper Section 5	ESTABLISHED IN MANUSCRIPT; exact endpoint evaluation
N5	Adjusted 98,306-ray fan is feasible with positive exact margin	Fan NPZ, orbit TSV, exact verifier	EXACT-CERTIFIED; NUMERICAL AUDIT; MUTATION-TESTED
N6	Forced-core volume lower bound	Exact fan volume and determinant transfer	EXACT-CERTIFIED; NUMERICAL AUDIT
N7	Three axis-specific antipodal cap-pair bounds	Hull TSV and cap certificate fields	EXACT-CERTIFIED; NUMERICAL AUDIT; MUTATION-TESTED
N8	Six cap interiors are pairwise disjoint	Paper and exact disjointness fields	ESTABLISHED IN MANUSCRIPT; EXACT-CERTIFIED; MUTATION-TESTED
N9	Near-Jung lower bound exceeds theorem upper enclosure	Final JSON assembly	EXACT-CERTIFIED; NUMERICAL AUDIT
T1	Universal coefficient is $26220940089713\pi/200000000000000$	03B02 plus strict near-Jung	EXACT-CERTIFIED; RECONSTRUCTION-CHECKED
T2	Scaling gives the theorem for arbitrary width $d > 0$	Homogeneity	ESTABLISHED IN MANUSCRIPT

Verification and provenance claims

ID	Claim	Evidence	Status
V1	Standalone verifier has no private repository dependency	Path regression test and source inspection	EXACT-CERTIFIED
V2	Final JSON is rebuilt byte identically	Exact verifier and unit test	RECONSTRUCTION-CHECKED
V3	Exact TSV and CSV renderings are rebuilt byte identically	Exact verifier	RECONSTRUCTION-CHECKED
V4	Binary64 implementation agrees on principal quantities	Binary64 transcript	NUMERICAL AUDIT
V5	Base-R implementation agrees on principal quantities	User-machine R transcript	NUMERICAL AUDIT
V6	Seven named proof-object mutations are rejected	Mutation transcript	MUTATION-TESTED
V7	Paper-facing constants and lower table match the JSON	Paper reconciliation script	RECONSTRUCTION-CHECKED
V8	Exact-constants appendix matches the JSON	Generator <code>--check</code>	RECONSTRUCTION-CHECKED
V9	Required release documents exist and preserve claim language	Release-document checker	RECONSTRUCTION-CHECKED

ID	Claim	Evidence	Status
V10	Markdown/PDF pairs are complete	Pair checker	RECONSTRUCTION-CHECKED
V11	PDFs are readable, have embedded fonts, and expose text	Poppler preflight	Passed in construction; hosted rerun pending

Publication and review claims

ID	Claim
P1	Lean nine-page Paper v1 manuscript is present
P2	Complete release-document set is present in Markdown and PDF
P3	Final author list and affiliation are approved
P4	License and citation metadata are final
P5	Public remote, immutable tag, and release assets exist
P6	Hosted Python, R, and document jobs are green on the release candidate
P7	Fresh public-clone replay is recorded
P8	External subject-matter reviewer has checked the theorem
P9	Peer review or journal acceptance exists

Claim boundary

The package does not establish equality, rigidity, sharpness, a minimizing body, or Meissner extremality. It should not be described as independently verified or formally verified while **P8** remains open.

Current bottleneck

The mathematical and finite-certificate package is internally replayed. The remaining release bottlenecks are administrative and external:

1. final authorship, affiliation, license, and citation metadata;
2. public remote and release-candidate commit;
3. green hosted workflow on that exact commit;
4. fresh public-clone replay;
5. user approval of the final paper, README, tag, and announcement; and
6. external mathematical review after release.